Let $B(x,y)$ be a rational function $x \in \mathbb{R}^n$ $y \in \mathbb{R}^m$

Let $\ell : \mathbb{R}^n \to \mathbb{R}^k$ be a linear map

Let B_i be the sum of all terms A in B s.t.

$$A(\lambda x, y) = \lambda^i A(x,y) \qquad i \in \mathbb{Z}, \quad \lambda \neq 0$$

$$\Rightarrow \quad B = \sum_i B_i \qquad \text{(finitely many terms)}$$

<u>Prop.</u> The following are equivalent:

① $\ell(x) = 0 \Rightarrow B(x,y) = 0$

② $\ell(x) = 0 \Rightarrow B_i(x,y) = 0 \quad \forall i$

<u>proof:</u> ② $\Rightarrow$ ① trivial.

Assume ① is true.

if $\ell(x) = 0$ and $\lambda \neq 0$ is arbitrary, then $\ell(\lambda x) = 0$

but then $B(\lambda x, y) = 0$ by assumption.

$$B = \sum_i B_i \qquad \text{so} \qquad B(\lambda x, y) = \sum_i B_i(\lambda x, y)$$

$$= \sum_i \lambda^i B_i(x,y) \qquad \text{Laurent polynomial in } \lambda$$

since λ is arbitrary, $f(\lambda) = \sum_i \lambda^i B_i(x,y) = 0$ implies that each coefficient of λ^i is zero.

$\boxed{\text{Connection with Radine}}$

variables X_{ij} are split in two

$X_{13}\ X_{14} \cdots X_{1n-1}\ X_{24}\ X_{25} \cdots X_{2n}$ $\rightarrow$ become the x variable

all others become y variable.

1 - zero: $\quad X_{2j} = X_{1j} - X_{13}$

think of it as
$$\begin{pmatrix} X_{24} - X_{14} + X_{13} \\ X_{25} - X_{15} + X_{13} \\ \vdots \\ X_{2n} + X_{13} \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ \vdots \end{pmatrix}$$

$$\ell(X_{13}, \dots, X_{24}, \dots) = \begin{pmatrix} X_{24} - X_{14} + X_{13} \\ X_{25} - X_{15} + X_{13} \\ \vdots \\ X_{2n} + X_{13} \end{pmatrix}$$

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for BCFW shift:
$$X_{1j}(z) = x_{1j}(0) + X^{\infty}_{1j}\, z$$
$$X_{2j}(z) = x_{2j}(0) + X^{\infty}_{2j}\, z$$
$$X_{ij}(z) = X_{ij}(0) \quad \text{if} \quad i,j \notin \{1,2\}$$

the x variable from the beginning satisfies $x(z) = x(0) + x^{\infty} z$

suppose $A(x,y)$ satisfies $A(\lambda x, y) = \lambda^{i} A(x,y)$

then $A(x(z), y) = A(x(0) + x^{\infty} z, y) \sim z^{i} A(x^{\infty}, y)$

$$\lim_{z\to\infty} \frac{A(x(0) + x^{\infty} z, y)}{z^{i} A(x^{\infty}, y)} = \lim_{z\to\infty} \frac{A(x(0) + x^{\infty} z, y)}{A(x^{\infty} z, y)} \overset{?}{=} \lim_{z\to\infty} A\left(\frac{x(0) + x^{\infty} z}{x^{\infty} z}, 1\right)$$

$$A(x,y) = \alpha\, x_1^{a_1} x_2^{a_2} \cdots x_n^{a_n}\, y_1^{b_1} \cdots y_m^{b_m}$$

$$\frac{A(x(0) + x^{\infty} z, y)}{A(x^{\infty} z, y)} = \frac{(x_1(0) + x^{\infty} z)^{a_1} \cdots (x_n(0) + x^{\infty} z)^{a_n}}{x_1^{\infty} z^{a_1} \cdots x_n^{\infty} z^{a_n}}$$

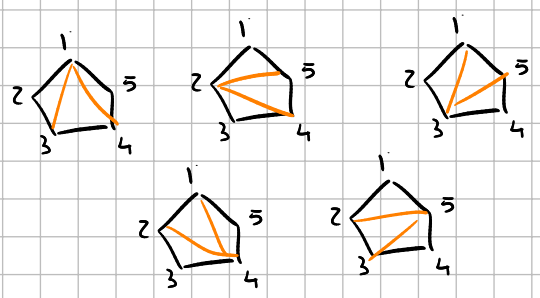
$\to 1 \qquad \to 1$

<u>5-point function</u>

X_{13}	X_{14}	X_{24}	X_{25}	X_{35}
x_1	x_2	x_3	x_4	y

$$B = \frac{a_1}{X_{13} X_{14}} + \frac{a_2}{X_{24} X_{25}} + \frac{a_3}{X_{13} X_{35}} + \frac{a_4}{X_{14} X_{24}} + \frac{a_5}{X_{25} X_{35}}$$

$$= \frac{a_1}{x_1 x_2} + \frac{a_2}{x_3 x_4} + \frac{a_3}{x_1 y} + \frac{a_4}{x_2 x_3} + \frac{a_5}{x_4 y}$$



$B_i = ?$

① <u>Rodine</u>

$$X_{1j}(z) = X_{1j}(0) + z X_{1j}^{\infty} \qquad X_{35}(z) = X_{35}^{\infty}$$

$$X_{2j}(z) = X_{2j}(0) + z X_{2j}^{\infty}$$

$$\frac{a_1}{X_{13}(z) X_{14}(z)} = \frac{a_1}{(X_{13}(0) + z X_{13}^{\infty})(X_{14}(0) + z X_{14}^{\infty})} \sim \frac{a_1}{z^2 X_{13}^{\infty} X_{14}^{\infty}}$$

$$\frac{a_2}{X_{24}(z) X_{25}(z)} = \cdots \sim \frac{a_2}{z^2 X_{24}^{\infty} X_{25}^{\infty}}$$

$$\frac{a_4}{X_{14} X_{24}} \sim \frac{a_4}{z^2 X_{14}^{\infty} X_{25}^{\infty}}$$

B_{-2}

$$\frac{a_3}{X_{13} X_{35}} \sim \frac{1}{z} \frac{a_3}{X_{13}^{\infty} X_{35}^{\infty}}$$

$$\frac{a_5}{X_{25} X_{35}} \sim \frac{1}{z} \frac{a_5}{X_{25}^{\infty} X_{35}^{\infty}}$$

B_{-1}

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②

$$\frac{a_1}{x_1 x_2} + \frac{a_2}{x_3 x_4} + \frac{a_4}{x_2 x_3}$$

B_{-2}

solo lim λ^{-2} if $x \to \lambda x$

$$\frac{a_3}{x_1 y} + \frac{a_5}{x_4 y}$$

B_{-1}

solo lim λ^{-1} if $x \to \lambda x$

$\boxed{\text{Multi-index notation}}$ $\quad x \in \mathbb{R}^n \quad \alpha \in \mathbb{N}_0^n$

$$x^{\alpha} \equiv x_1^{\alpha_1} x_2^{\alpha_2} \cdots x_n^{\alpha_n} \qquad |\alpha| = \alpha_1 + \alpha_2 + \cdots + \alpha_n$$

Assuming B is homogeneous and all powers of x, y are negative

$$\boxed{B(x,y) = \sum_{\substack{\alpha \in \mathbb{N}_0^n \\ \beta \in \mathbb{N}_0^m \\ |\alpha| + |\beta| = N}} \frac{a_{\alpha\beta}}{x^{\alpha} y^{\beta}}}$$

something eq. to D-subsets

suppose $\ell(x)=0 \Rightarrow B(x,y)=0$. Then $\ell(x)=0 \Rightarrow B_{-k}(x,y)=0 \ \forall k$

$$B_{-k}(x,y) = \sum_{\substack{\alpha\in\mathbb{N}_0^n \\ \beta\in\mathbb{N}_0^m \\ |\alpha|=k \\ |\beta|=N-k}} \frac{a_{\alpha\beta}}{x^\alpha y^\beta}$$

→ essentially D-subsets

since the y is arbitrary it must be

$$\ell(x)=0 \Rightarrow \sum_{\substack{\alpha\in\mathbb{N}_0^n \\ |\alpha|=k}} \frac{a_{\alpha\beta}}{x^\alpha} = 0 \qquad \forall\ \beta\in\mathbb{N}_0^m \text{ s.t } |\beta|=N-k$$

X_{13} X_{14} X_{24} X_{25} X_{35} $\qquad N=2 \qquad k=0,1,2$

x_1 x_2 x_3 x_4 y

$$B(x,y) = \underbrace{\left|\frac{a_1}{x_1x_2}+\frac{a_2}{x_1x_3}+\frac{a_3}{x_1x_4}+\frac{a_4}{x_2x_3}+\frac{a_5}{x_2x_4}+\frac{a_6}{x_3x_4}\right|}_{B_{-2}} + \underbrace{\left|\frac{a_7}{x_1y}+\frac{a_8}{x_2y}+\frac{a_9}{x_3y}+\frac{a_{10}}{x_4y}\right|}_{B_{-1}} + \underbrace{\left|\frac{a_{11}}{yz}\right|}_{B_0}$$

$\ell(x)=0 \Rightarrow B=0$

$$\boxed{\begin{aligned} x_3 &= x_2 - x_1 \\ x_4 &= -x_1 \end{aligned}} \qquad \left(\begin{aligned} X_{24} &= X_{14} - X_{13} \\ X_{25} &= -X_{13} \end{aligned}\right) \quad (1\text{-zero})$$

forget $\frac{1}{x_1^2}, \frac{1}{x_2^2}, \frac{1}{x_3^2}, \frac{1}{x_4^2}$

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look at B_{-1} with constraint

$$B_{-1}=0 \Leftrightarrow a_7 x_2x_3x_4 + \cancel{a_8}\, x_1x_3x_4 + \cancel{a_9}\, x_1x_2x_4 + a_{10}\, x_1x_2x_3 = 0$$

now impose constraint

→ get a polynomial in x_1, x_2

$$a_7 x_2(x_2-x_1)(-x_1) + a_9 x_1(x_2-x_1)(-x_1) + - \quad - \quad = 0$$

all coefficients → of powers of $x_1 x_2$ must be zero

coefficient of x_1^3 is zero
" " $x_1^2 x_2$ is zero
$x_1 x_2^2$ is zero

coeff. of x_1^3 $\rightarrow$ $\boxed{a_8 = 0}$

coeff. of $x_1^2 x_2$ $\rightarrow$ $\boxed{a_9 = 0}$

left with $a_7 \boxed{x_2(x_2 - x_1)(-x_1)} + a_{10} \boxed{x_1 x_2 (x_2 - x_1)}$

$= (a_7 - a_{10})(\ - - \ ) = 0 \rightarrow \boxed{a_7 = a_{10}}$

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idea want to show this

$$\ell(x) = 0 \Rightarrow \sum_{\substack{\alpha \in \mathbb{N}_0^n \\ |\alpha| = k}} \frac{a_{\alpha\beta}}{x^\alpha} = 0 \quad \forall \beta \in \mathbb{N}_0^m \text{ s.t } |\beta| = N - k$$

1-zero for n legs:

$$X_{24} = X_{14} - X_{13}$$
$$X_{25} = X_{15} - X_{13}$$
$$\vdots$$
$$X_{2n-1} = X_{1n-1} - X_{13}$$
$$X_{2n} = - X_{13}$$

1-zero for $n+1$ legs:

$$X_{24} = X_{14} - X_{13}$$
$$X_{25} = X_{15} - X_{13}$$
$$\vdots$$
$$X_{2n-1} = X_{1n-1} - X_{13}$$
$$X_{2n} = X_{1n} - X_{13}$$
$$X_{2n+1} = - X_{13}$$

$x_1 = x_{13}$ $\quad x_2 = x_{24}$

$n=4$ $\quad \frac{a}{x_1} + \frac{b}{x_4} = 0$ when $x_4 = -x_1 \Rightarrow a = b$

$N=1$

$n=5 \quad N=2$

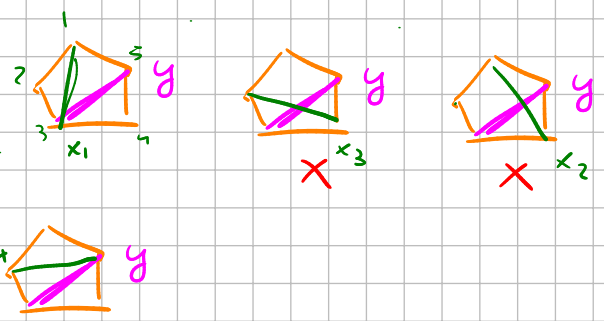
$$\frac{1}{y}\left(\frac{a_7}{x_1} + \frac{a_{10}}{x_4} + \frac{a_8}{x_2} + \frac{a_9}{x_3}\right)$$

$$\frac{a_7}{x_1 y} + \frac{a_8}{x_2 y} + \frac{a_9}{x_3 y} + \frac{a_{10}}{x_4 y} = 0$$

when $\boxed{\begin{array}{l} x_3 = x_2 - x_1 \\ x_4 = -x_1 \end{array}}$

intersecting chords

now that



alternative approach

look at growth when x_1, x_2 do things

$x_1\ x_2$ independent

- $x_1 \to \infty$ $\quad x_2$ fixed $\quad$ so $\quad x_3 \to -\infty$, $x_4 \to -\infty$ $\quad 0 = B \Rightarrow \frac{a_8}{x_2} = 0$ $\quad$ so $a_8 = 0$

$x_3\ x_4$ independent

- $x_4 \to \infty$ $\quad x_3$ fixed $\quad x_2 \to \infty$, $x_1 \to \infty$ $\quad 0 = B \Rightarrow \frac{a_9}{x_3} = 0$ $\quad$ so $a_9 = 0$

$$\underbrace{\boxed{\frac{a_1}{x_1 x_2} + \frac{a_2}{x_1 x_3} + \frac{a_3}{x_1 x_4}} + \boxed{\frac{a_4}{x_2 x_3} + \frac{a_5}{x_2 x_4} + \frac{a_6}{x_3 x_4}}}_{=0} + \frac{b_1}{x_1^2} + \frac{b_2}{x_2^2} + \frac{b_3}{x_3^2} + \frac{b_4}{x_4^2} = 0$$

$x_3 = x_2 - x_1$

$x_4 = -x_1$

$x_1 \to \infty$, x_2 fixed, $x_3 \to \infty$, $x_4 \to \infty$ $\quad \to \quad b_2 = 0$

x_3 fixed, $x_1, x_2, x_4 \to \infty$ $\quad \to \quad \frac{b_3}{x_3^2} = 0 \to b_3 = 0$

x_1 fixed, x_4 fixed, $x_2, x_3 \to \infty$ $\quad \to \quad \frac{1}{x_1^2}\left(\underbrace{\boxed{b_1 + b_4 - a_3}}_{=0}\right) = 0$

when $x_4 = -x_1$

$$\frac{a_3}{x_1 x_4} + \frac{b_1}{x_1^2} + \frac{b_4}{x_4^2} = 0$$

$$\frac{a_1}{x_1 x_2} + \frac{a_2}{x_1 x_3} + \frac{a_4}{x_2 x_3} + \frac{a_5}{x_2 x_4} + \frac{a_6}{x_3 x_4} = 0$$

common denominator:

$$a_1 x_3 x_4 + a_2 x_2 x_4 + a_4 x_1 x_4 + a_5 x_1 x_3 + a_6 x_1 x_2 = 0$$

$x_4 = -x_1$
$x_3 = x_2 - x_1$

coefficient of x_1^2: $a_1 = a_4 + a_5$

coefficient of x_2^2:

coefficient of $x_1 x_2$ $a_2 = a_6 - a_4$

summarize

$$\begin{cases} b_2 = b_3 = a_{11} = a_8 = a_9 = 0 \\ a_7 = a_{10} \\ a_1 = a_4 + a_5 \\ a_2 = a_6 - a_4 \\ b_1 + b_4 - a_3 = 0 \end{cases}$$

$$B(x,y) = \frac{a_1}{x_1 x_2} + \frac{a_2}{x_1 x_3} + \frac{a_3}{x_1 x_4} + \frac{a_4}{x_2 x_3} + \frac{a_5}{x_2 x_4} + \frac{a_6}{x_3 x_4} + \frac{a_7}{x_1 y} + \frac{a_8}{x_2 y} + \frac{a_9}{x_3 y} + \frac{a_{10}}{x_4 y} + \frac{a_{11}}{y^2}$$

$$+ \frac{b_1}{x_1^2} + \frac{b_2}{x_2^2} + \frac{b_3}{x_3^2} + \frac{b_4}{x_4^2}$$

$\downarrow$

$$B(x,y) = \frac{a_4 + a_5}{x_1 x_2} + \frac{a_6 - a_4}{x_1 x_3} + \frac{b_1 + b_4}{x_1 x_4} + \frac{\bar{a}_4}{x_2 x_3} + \frac{\bar{a}_5}{x_2 x_4} + \frac{\bar{a}_6}{x_3 x_4} + \frac{\bar{a}_7}{y}\left(\frac{1}{x_1} + \frac{1}{x_4}\right)$$

$$+ \frac{\bar{b}_1}{x_2^2} + \frac{\bar{b}_4}{x_4^2}$$

impose other constraint (cyclic shift $X_{24} = X_{14} - X_{13}$ $\rightarrow$ $X_{35} = X_{25} - X_{24}$

$X_{25} = -X_{13}$ $\quad X_{31} = -X_{24}$

$y = x_4 - x_3$
$x_1 = -x_3$

$$\boxed{\begin{array}{lll} x_1 = X_{13} & x_2 = X_{14} & \\ x_3 = X_{24} & x_4 = X_{25} & y = X_{35} \end{array}}$$

$$B = \frac{a_4 + \cancel{a_5}}{x_1 x_2} + \frac{a_6 - a_4}{x_1 x_3} + \frac{b_1 + b_4}{x_1 x_4} + \frac{a_4}{x_2 x_3} + \cancel{\frac{a_5}{x_2 x_4}} + \frac{a_6}{x_3 x_4} + \frac{a_7}{y}\left(\frac{1}{x_1} + \frac{1}{x_4}\right) + \frac{b_1}{x_1^2} + \frac{b_4}{x_4^2}$$

$$\boxed{\begin{aligned} y &= x_4 - x_3 \\ x_1 &= -x_3 \end{aligned}}$$

(1) x_2 is arbitrary $\qquad \frac{a_4 + a_5}{x_1} + \frac{a_4}{x_3} + \frac{a_5}{x_4} = 0$

treat x_3, x_4 as arbitrary: $\qquad \frac{-a_4 - a_5 + a_4}{x_3} + \frac{a_5}{x_4} = 0 \quad \forall\, x_4, x_3$

$a_5 = 0$

$$B = \frac{a_4}{x_1 x_2} + \frac{a_6 - a_4}{x_1 x_3} + \frac{b_1 + b_4}{x_1 x_4} + \frac{a_4}{x_2 x_3} + \frac{a_6}{x_3 x_4} + \frac{a_7}{y}\left(\frac{1}{x_1} + \frac{1}{x_4}\right) + \frac{b_1}{x_1^2} + \frac{b_4}{x_4^2}$$

(2) $\frac{a_6 - a_4}{x_1 x_3} + \frac{b_1 + \cancel{b_4}}{x_1 x_4} + \frac{a_6}{x_3 x_4} + \frac{a_7}{y}\left(\frac{1}{x_1} + \frac{1}{x_4}\right) + \frac{b_1}{x_1^2} + \frac{\cancel{b_4}}{x_4^2} = 0 \qquad \begin{cases} y = x_4 - x_3 \\ x_1 = -x_3 \end{cases}$

- $x_3 \to \infty$, x_4 fixed, $y \to \infty$, $x_1 \to \infty$ $\quad \to \quad b_4 = 0$

- x_1, x_3 fixed, $x_4, y \to \infty$ $\qquad \frac{b_1}{x_1^2} - \frac{a_6 - a_4}{x_1^2} = 0 \qquad \boxed{b_1 = a_6 - a_4}$

$$\hookrightarrow \quad \frac{b_1}{x_1 x_3} + \frac{b_1}{x_1^2} + \frac{b_1}{x_1 x_4} + \frac{a_6}{x_3 x_4} + \frac{a_7}{x_1 y} + \frac{a_7}{x_4 y}$$

common denominator:

$$b_1\left(x_1 x_4 y + x_3 x_4 y + x_1 x_3 y\right) + a_6\, x_1^2 y + a_7\left(x_1 x_3 x_4 + x_1^2 x_3\right)$$

$x_3^3: \quad \propto \quad b_1 + a_7 - a_6 = 0$

$x_3^2 x_4: \quad \propto \quad a_6 - a_7 - b_1 = 0$

$y = x_4 - x_3$

$x_1 = -x_3$

$$\boxed{a_6 = b_1 + a_7} \qquad \boxed{b_1 = a_6 - a_4}$$

$$B = \frac{a_4}{x_1 x_2} + \frac{b_1}{x_1 x_3} + \frac{b_1}{x_1 x_4} + \frac{a_4}{x_2 x_3} + \frac{a_6}{x_3 x_4} + \frac{a_7}{y}\left(\frac{1}{x_1} + \frac{1}{x_4}\right) + \frac{b_1}{x_2^2}$$

impose 1st 1-zero